

MODUL 3

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Soal Latian

1.Kode Program

```
package main import
"fmt"

func factorial(n int) int {
    res := 1    for i := 2; i <=
n; i++ {      res *= i
    }
    return res
}

func permutation(n, r int) int {
    return factorial(n) / factorial(n-r)
}

func combination(n, r int) int {    return
factorial(n) / (factorial(r) * factorial(n-r))
}
```

```

func main() {

    var a, b, c, d int

    fmt.Scan(&a, &b, &c, &d)

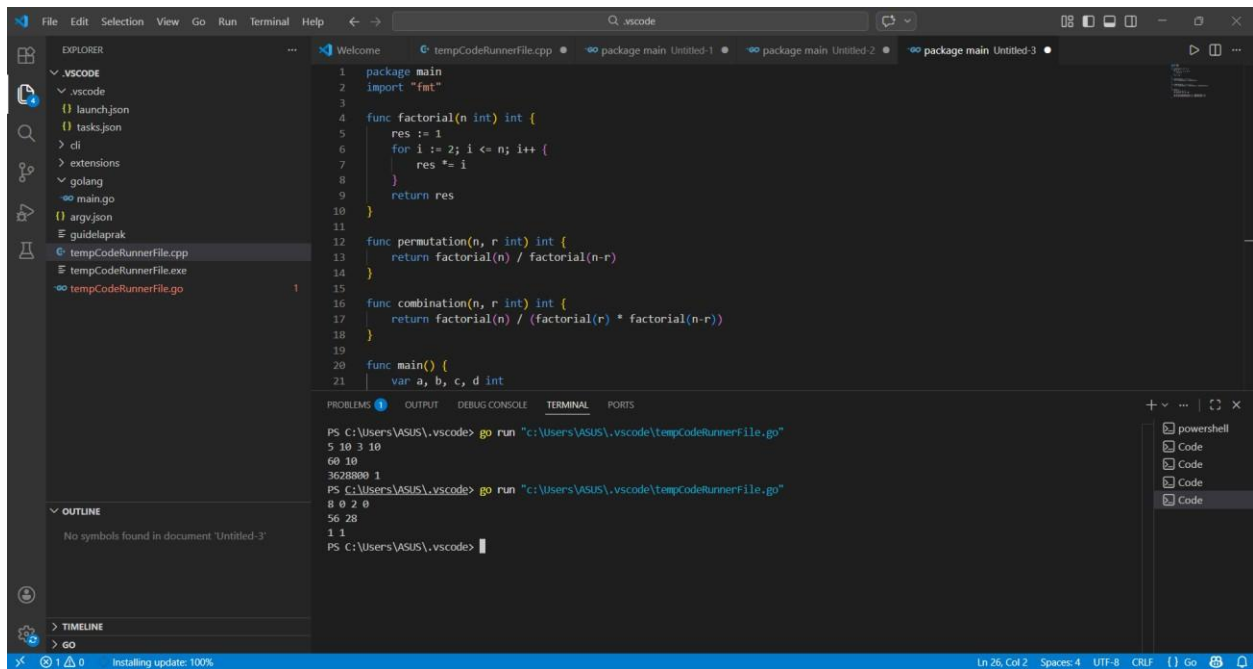

    fmt.Println(permutation(a, c), combination(a, c))

    fmt.Println(permutation(b, d), combination(b, d))

}

```

Screenshot Program



2.Kode Program

```

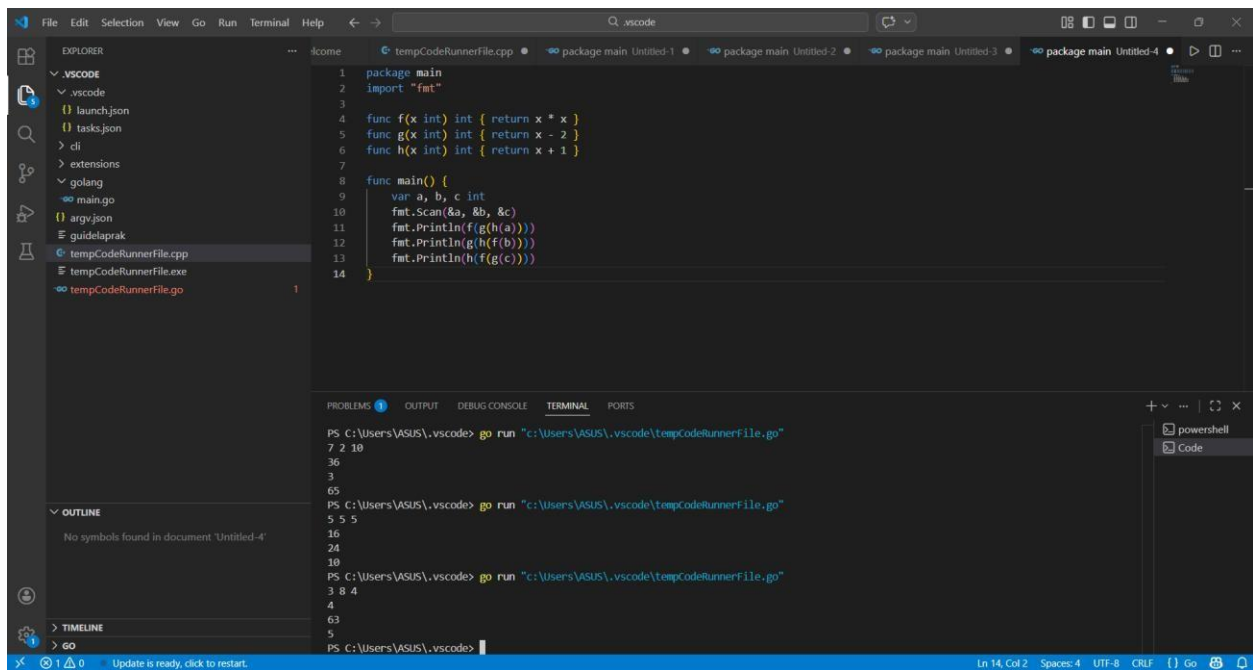
package main import
"fmt" func f(x int) int
{ return x * x } func
g(x int) int { return x

```

```
- 2 } func h(x int) int {  
return x + 1 }
```

```
func main() {  
    var a, b, c int  
    fmt.Scan(&a, &b, &c)  
    fmt.Println(f(g(h(a))))  
    fmt.Println(g(h(f(b))))  
    fmt.Println(h(f(g(c))))  
}
```

Screenshot Program



3.Kode Program

```
package main import
(
    "fmt"
    "math"
)

func jarak(a, b, c, d float64) float64 {    return
math.Sqrt(math.Pow(a-c, 2) + math.Pow(b-d, 2))
}

func didalam(cx, cy, r, x, y float64) bool {
return jarak(cx, cy, x, y) <= r
}

func main() {    var cx1,
cy1, r1 float64    var cx2,
cy2, r2 float64    var x, y
float64

    fmt.Scan(&cx1, &cy1, &r1)
fmt.Scan(&cx2, &cy2, &r2)
fmt.Scan(&x, &y)

    in1 := didalam(cx1, cy1, r1, x, y)
in2 := didalam(cx2, cy2, r2, x, y)
```

```

    if in1 && in2 {      fmt.Println("Titik di dalam
lingkaran 1 dan 2")

    } else if in1 {      fmt.Println("Titik di
dalam lingkaran 1")

    } else if in2 {      fmt.Println("Titik di
dalam lingkaran 2")

    } else {      fmt.Println("Titik di luar lingkaran
1 dan 2")

    }

}

```

Screenshot Program

```

15 func main() {
16     var x, y float64
17
18     fmt.Scan(&cx1, &cy1, &r1)
19     fmt.Scan(&cx2, &cy2, &r2)
20     fmt.Scan(&x, &y)
21
22     in1 := didalam(cx1, cy1, r1, x, y)
23     in2 := didalam(cx2, cy2, r2, x, y)
24
25     if in1 && in2 {
26         fmt.Println("Titik di dalam lingkaran 1 dan 2")
27     } else if in1 {
28         fmt.Println("Titik di dalam lingkaran 1")
29     } else if in2 {
30         fmt.Println("Titik di dalam lingkaran 2")
31     } else {
32         fmt.Println("Titik di luar lingkaran 1 dan 2")
33     }
34 }
35
36

```

Terminal Output:

```

PS C:\Users\ASUS\.vscode> go run "c:\Users\ASUS\.vscode\tempCodeRunnerFile.go"
1 1 5
8 8 4
2 2
Titik di dalam lingkaran 1
PS C:\Users\ASUS\.vscode> go run "c:\Users\ASUS\.vscode\tempCodeRunnerFile.go"
1 2 3
4 5 6
7 8
Titik di dalam lingkaran 2
PS C:\Users\ASUS\.vscode> go run "c:\Users\ASUS\.vscode\tempCodeRunnerFile.go"
5 10 15
-15 4 20
0 0
Titik di dalam lingkaran 1 dan 2
PS C:\Users\ASUS\.vscode>

```

